1. Prove that there do not exist differentiable functions f and g for which $f(0) = g(0) = 0$ and $x = f(x)g(x)$.

2. Suppose $f : \mathbb{R} \to \mathbb{R}$ is differentiable and define $g(x) = x^2 f(x^3)$. Show that g is differentiable and compute g'.

3. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = \dfrac{1}{1+x^2}$. Prove that f has a maximum value and find the point at which the maximum occurs.

4. Show that $\cos(x) = x^3 + x^2 + 4x$ has exactly one root in $\left[0, \frac{\pi}{2}\right]$.

5. Suppose $f : [0, 2] \to \mathbb{R}$ is differentiable, $f(0) = 0$, $f(1) = 2$ and $f(2) = 2$. Prove that
   (a) there is c_1 such that $f'(c_1) = 0$,
   (b) there is c_2 such that $f'(c_2) = 2$,
   (c) there is c_3 such that $f'(c_3) = \frac{3}{2}$.

6. Use the Mean-Value Theorem to prove that
$$ny^{n-1}(x-y) \le x^n - y^n \le nx^{n-1}(x-y)$$
   if $n \ge 1$ and $0 \le y \le x$.

7. Use the Mean-Value Theorem to prove that
$$\sqrt{1+h} < 1 + \frac{1}{2}h \text{ for } h > 0.$$

8. Show that if $0 < p < 1$ and $h > 0$, then
$$(1+h)^p < 1 + ph.$$